

Evolved biological factories: constraint-based evolutionary search for robust metabolic architectures in *E. coli*

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1 Introduction

Engineered microbes convert renewable feedstocks into fuels, materials, and chemicals, but the design of production strains remains guided primarily by human intuition: an engineer picks a pathway, expresses it, and hopes the host tolerates it. An alternative is to let a search process explore the space of possible metabolic architectures and select for what works, under conditions that resemble an industrial process rather than a single idealized optimum. This project asks whether a computational evolutionary search, starting from a target product rather than an existing pathway, can discover non-obvious biological architectures that match or exceed known engineered pathways under multi-condition robustness criteria. The design principles that let engineered cells maintain high productivity while remaining metabolically stable are the scientific object of interest; the evolved architecture library is the deliverable.

Three hypotheses are pre-registered (PRE-REGISTRATION.md, locked before any data or search): H1 (performance) - at least one evolved architecture reaches predicted product flux at or above the published engineered pathway evaluated in the same genome-scale model under identical uptake bounds; H2 (novelty) - the evolved library contains architectures absent from the published pathway that are mechanistically interpretable; H3 (robustness principle) - architectures selected under multi-condition pressure retain their production across a locked stress battery at a significantly higher rate than architectures selected under single-condition pressure. Honest negatives are preserved and redirect the project internally; no negative is published as the final result.

2 Pre-registered design

2.1 Hosts, models, and targets

The primary host is *Escherichia coli* K-12 MG1655 in the genome-scale metabolic model iML1515, with the yeast arm (Yeast8) reserved for cross-host robustness analysis. Three locked target products span distinct chemistry classes: 1,4-butanediol (non-natural industrial diol; benchmark route Yim et al. 2011, the Genomatica pathway), isobutanol (branched-chain higher alcohol; benchmark Atsumi et al. 2008, the keto-acid route), and lycopene (tetraterpene; benchmark Alper et al. 2005, crtEBI with dxs push). Reaction lists per the cited papers were locked in data/benchmarks.json before any search, with DOI provenance.

2.2 Benchmark metric: production envelopes

For each target, the published engineered pathway was implemented in iML1515 under the locked base condition (M9 minimal, glucose 10 mmol/gDW/h, aerobic). Because production competes with biomass, a single max-growth number is meaningless: at 100% of maximum growth all three pathways carry zero product flux. The benchmark is therefore a production envelope: maximum product flux with growth fixed at locked fractions of the FBA maximum,

$$F_t(g) = \max_v v_{\text{product}} \quad \text{s.t.} \quad Sv = 0, v_{\text{biomass}} = g \cdot \mu_{\text{max}}, l \leq v \leq u, \quad (1)$$

evaluated at $g \in \{1.0, 0.9, 0.75, 0.5, 0.25, 0.0\}$ with a parsimonious-FBA tie-break. Measured envelopes (results/benchmark_flux.json): 1,4-BDO reaches $F = 10.35$ mmol/gDW/h at $g = 0$ and 5.27 at $g = 0.5$; isobutanol 10.00 and 5.24; lycopene 0.97 and 0.49. The envelope, not a single titer, is what an evolved architecture must beat at matched growth. Table 1 reports the measured envelopes for all three published routes at the six locked growth fractions; the FBA maximum growth rate of the host under the base condition is $\mu_{\text{max}} = 0.877 \text{ h}^{-1}$ for all targets (identical uptake bounds, wild-type core). The zero at $g = 1.0$ in every row is a verified property

Table 1: Measured production envelopes (product flux, mmol/gDW/h) of the three published engineered routes in iML1515 under the locked base condition, at growth fixed to fractions of μ_{max} (results/benchmark_flux.json, run 2026-09-26; pFBA tie-break coefficient 0.999).

Target route	$g=1.0$	0.9	0.75	0.5	0.25	0.0
1,4-BDO (Yim 2011)	0.000	1.060	2.649	5.267	7.878	10.355
Isobutanol (Atsumi 2008)	0.000	1.048	2.620	5.239	7.813	10.000
Lycopene (Alper 2005)	0.000	0.098	0.244	0.488	0.731	0.970

of the host–route system, not a solver artifact: biomass and product draw on the same carbon and redox budget, so the envelopes are strictly monotone decreasing in g , and any comparison at a single growth point would be meaningless without the matched-growth rule.

2.3 Search protocol

The tie-break inside each envelope point is parsimonious FBA: among optimal solutions, minimize total flux,

$$v^* = \arg \min_v \|v\|_1 \quad \text{s.t.} \quad Sv = 0, v_{\text{product}} = F_t(g), v_{\text{biomass}} = g \mu_{\text{max}}, l \leq v \leq u, \quad (2)$$

with the pFBA optimality slack fixed at 0.999; FBA and pFBA product fluxes were verified equal on all envelope points (unique optima), so the cheaper FBA is used for search fitness and full pFBA reserved for final evaluation. The search genome encodes on/off choices over a locked reaction pool (benchmark route blocks plus literature-noted alternatives, data/reaction_pool.json). Fitness is product flux at the pFBA-optimal solution with growth fixed at $g = 0.5$, the envelope midpoint. Search fitness is the nominal-condition envelope point at the locked growth fraction,

$$\Phi(a) = F_a^{(\text{nominal})}(0.5), \quad (3)$$

and architecture performance against the benchmark route is reported as the exceedance ratio

$$E_a = \frac{F_a^{(\text{nominal})}(0.5)}{F_{\text{benchmark}}^{(\text{nominal})}(0.5)}, \quad (4)$$

with $E_a > 1$ required for an H1 beat at matched uptake bounds. The exhaustive pools used so far have genome spaces of

$$|\mathcal{G}| = 2^h \times 2^t, \quad (5)$$

genomes per target (h host-level blocks, t target-specific blocks: $h=0, t=2$ in pool v1 giving 4; $h=4, t=2$ in pool v2 giving 64), so enumeration evaluates every genome exactly once and the stochastic search budget is dominated by construction. Significance of any evolved advantage is calibrated against a 1000-seed random-architecture null as the percentile

$$\hat{q} = \frac{1 + \sum_{s=1}^{1000} \mathbf{1}[\Phi(a_s^{\text{rand}}) \geq \Phi(a)]}{1001}. \quad (6)$$

The search is a $(\mu + \lambda)$ evolutionary algorithm ($\mu = 24$, $\lambda = 48$, tournament selection, 60 generations, five locked seeds), with a 1000-seed random-architecture null for significance calibration. On the version-1 pool (two alternative blocks per target, four genomes), exhaustive enumeration strictly dominates the stochastic search and was used instead; the EA scaffold is retained for expanded pools.

2.4 Stability battery

A scalar summary of production-growth coupling used in the envelope analysis is the local slope

$$\sigma_t(g) = \frac{F_t(g_2) - F_t(g_1)}{g_2 - g_1}, \quad (7)$$

the shadow price of growth in units of product flux per unit growth fraction; the measured envelopes are strictly monotone with $\sigma_t < 0$ throughout. Robustness is evaluated over a locked nine-condition battery (nutrient shifts, carbon shifts to glycerol and acetate, anaerobiosis, uptake perturbations, oxygen limitation). After a metric-interpretation bug was caught at first evaluation (growth referenced to the base-condition maximum, which made every condition untestable), the locked rule is growth fixed at 50% of the condition-specific maximum, with retention defined against nominal-condition flux. The nine locked conditions are: GLC_LOW (glucose uptake -2), GLC_MID (-5), GLC+20% (-12), GLC -20% (-8), GLYCEROL (sole carbon, uptake -10), ACETATE (sole carbon, uptake -10), ANAEROBIC (oxygen uptake 0), O2 -50% (oxygen uptake halved), and the nominal base GLC_AEROBIC (glucose -10 , aerobic); all uptakes in mmol/gDW/h. Formally, for architecture a and condition c ,

$$R(a, c) = \frac{F_a^{(c)}(0.5)}{F_a^{(\text{nominal})}(0.5)}, \quad \text{BatteryPass}(a) = \bigwedge_{c \in \text{battery}} \mathbf{1}[R(a, c) \geq 0.8], \quad (8)$$

where $F_a^{(c)}(0.5)$ is the architecture's product flux in condition c with growth fixed at 50% of the condition-specific maximum. The pass bar of 0.8 was locked before any architecture was evaluated against the battery.

3 Results

3.1 Single-condition search reproduces the benchmark envelopes

Across all fifteen seed runs (three targets, five seeds), the best evolved architecture exactly reaches the locked envelope value at $g = 0.5$: 5.27 mmol/gDW/h for 1,4-BDO, 5.24 for isobu-

tanol, 0.49 for lycopene. The search recovers the published routes as optimal under single-condition selection - a validation of the evaluation pipeline, and the baseline for H1.

3.2 Search definitions (locked terminology)

To remove any ambiguity about what the method is: the *representation* is a genome vector of discrete, biologically typed block choices (h host-level levers and t target-specific route blocks over the locked reaction library); the *search policy* is either exhaustive enumeration of the genome space (strictly dominant at pool sizes ≤ 256 and used for all reported results) or a $(\mu + \lambda)$ evolutionary algorithm ($\mu = 24$, $\lambda = 48$, tournament selection, 60 generations, five locked seeds) retained for larger pools; the *objective* combines nominal-condition product flux against the matched-growth published-route envelope and the tiered robustness score R_{tiered} over the locked nine-condition battery; the *learning component* is none - no trained, fitted, or adaptive model appears anywhere in the pipeline, and no parameter is updated from data. The method is constraint-based evolutionary search over a genome-scale stoichiometric model, nothing more. Where a future amendment adds a learned surrogate or policy, it will be locked and named explicitly.

3.3 H2 candidate: a tail-only minimal 1,4-BDO architecture

A non-obvious architecture emerged from the search: with *both* heterologous upstream route blocks switched off, the four-step heterologous tail (4-hydroxybutyrate dehydrogenase, CoA transferase, CoA-acylating aldehyde dehydrogenase, alcohol dehydrogenase) still reaches the full nominal flux of 5.27 mmol/gDW/h. The model explains why: *E. coli*'s native GABA shunt (succinate-semialdehyde supply through glutamate metabolism) feeds the tail without any heterologous upstream enzyme. This tail-only architecture is absent from the Yim et al. pathway definition, uses four fewer heterologous enzymes, and is mechanistically interpretable - the H2 novelty pattern in its locked form at the nominal condition. As the tiered-battery subsection below shows, the candidate does not survive the battery, so the claim stands at nominal condition only.

3.4 H3 honest negative on pool v1, and a battery-design finding

Exhaustive multi-condition evaluation of pool v1 found that *no* genome passes the locked battery (80% retention in all nine conditions): best minimum retentions are 0.175 (1,4-BDO), 0.176 (isobutanol), and 0.27 (lycopene). Inspection shows the failure is structural, not a lack of search power: under the severe nutrient shift GLC_LOW (glucose at 2 rather than 10 mmol/gDW/h), elementary carbon balance caps a C4 product at roughly one to two mmol/gDW/h against a nominal 5.27, a retention ceiling of 0.2–0.3 that no architecture can lift. The bound is elementary: a C4 product made from glucose at uptake u_{glc} satisfies

$$v_{\text{product}} \leq \frac{6}{4} u_{\text{glc}} = 3.0 \text{ mmol/gDW/h at } u_{\text{glc}} = 2, \quad (9)$$

before any carbon is allocated to biomass, maintenance, or byproducts; with growth fixed at 50% of the condition maximum the achievable product flux falls to roughly one to two, giving $R(a, \text{GLC_LOW}) \leq 1.5/5.267 \approx 0.28$ for every conceivable architecture a - including the published routes and any architecture the search could ever return. The benchmark engineered

routes fail identically. A pass bar of 80% retention on severe nutrient shifts is therefore unsatisfiable by construction, the battery provides no selection gradient, and H3 is untestable as locked. This is recorded as an honest negative with the design flaw named, and a redesign (retention against condition-achievable optimum rather than against nominal flux; tiered bars for shifts versus perturbations; robustness-oriented pool levers such as knockout blocks and co-utilization feeds) is being locked as a dated amendment before any re-evaluation, following the project’s redirect-on-failure rule.

3.5 Pool v2: robustness levers shift the nominal optimum (amendment 2026-09-26)

Following the redirect rule, pool v2 was locked as a dated amendment before evaluation: four host-level robustness levers - acetate-overflow knockout (ackA-pta), formate-overflow knockout (pflB), lactate-overflow knockout (ldhA), and a glycerol co-utilization feed - added to the target-specific blocks, expanding the genome space to 64 genomes per target. Exhaustive enumeration (192 genomes total, metric-independent of the pending battery redesign) was again strictly dominant over the stochastic budget at this size. Under the nominal condition at $g = 0.5$, the co-utilization feed dominates every target: best nominal fluxes rise to 6.496 (1,4-BDO), 6.498 (isobutanol), and 0.730 (lycopene) mmol/gDW/h, against envelope values of 5.267, 5.239, and 0.488 - nominal-condition exceedances of 23%, 24%, and 50% over the published routes evaluated in the same model. The gain is attributable to the added carbon supply, so the honest comparison under co-feeding must re-baseline the benchmark route with the same feed block enabled:

$$E_a^{\text{cofeed}} = \frac{F_a^{(\text{cofeed})}(0.5)}{F_{\text{benchmark}}^{(\text{cofeed})}(0.5)}, \quad (10)$$

with both sides evaluated under identical uptake bounds including the glycerol co-feed. That re-baselined comparison has now been executed (amendment 2026-09-27 10:04): $E^{\text{cofeed}} = 1.00$ on all three targets (6.496 vs 6.496; 6.498 vs 6.498; 0.730 vs 0.730) - the nominal exceedance is entirely attributable to the added carbon supply, and no nominal-flux advantage over the published routes survives equal-feed comparison. The 1000-draw random-architecture null likewise recovers the pool optimum on all three targets, confirming that nominal-condition search is trivial at this pool size; the project’s discriminating content is the robustness structure below, not nominal flux. An honest secondary negative is recorded: for isobutanol, some knockout combinations abolish product flux entirely (minimum 0.0), so the levers are not uniformly benign even at nominal condition.

3.6 Tiered battery executed: satisfiability restored, H3 advantage unearned (amendment 2026-09-27)

The battery was redesigned and locked as a dated amendment on 2026-09-27 before evaluation: mild perturbations (glucose $\pm 20\%$, oxygen -50%) are scored as absolute retention versus nominal with a hard leg of ≥ 0.8 each, and severe nutrient shifts (GLC_LOW, GLC_MID, ANAEROBIC, GLYCEROL, ACETATE) are scored against the pool frontier in the same condition, $r_k = v_{p,k} / \max_a v_{p,k}(a)$, with a hard leg of ≥ 0.8 each; $R_{\text{tiered}} = 0.5 R_{\text{pert}} + 0.5 R_{\text{shift}}$. The bar is now satisfiable: 16/64 (1,4-BDO), 18/64 (isobutanol), and 18/64 (lycopene) genomes pass both hard legs. The best designs score $R_{\text{tiered}} = 1.0$ in all three targets: pflB knockout

plus glycerol co-feed plus the succinyl-CoA route at 6.496 mmol/gDW/h nominal (1,4-BDO), pflB knockout plus co-feed plus the NADH alcohol dehydrogenase at 6.497 (isobutanol), and co-feed plus idi push at 0.730 (lycopene). Two findings follow. First, the tail-only H2 candidate does *not* survive the tiered battery ($R_{\text{tiered}} = 0.768$; severe-condition frontier-relative retention 0.41–0.67 on GLC_LOW, GLC_MID, ANAEROBIC, and ACETATE): its minimal-architecture claim stands at the nominal condition only, and the battery-robust designs are co-fed, not enzyme-minimal. Second, the H3 comparison is degenerate at this pool size: the best-by-nominal genome and the best-by- R_{tiered} genome coincide in all three targets (battery pass rates 3/3 versus 3/3), so no multi-condition search advantage is demonstrable at pool 64. This is recorded as an honest negative at current scale, not a refutation of the principle. The queued resolution was executed under the 2026-09-27 10:04 amendment: pool v3 (two further host levers, adhE and tpiA knockouts; 256 genomes per target, exhaustive) again yields $\text{argmax}(v_{\text{nominal}}) = \text{argmax}(R_{\text{tiered}})$ in all three targets (rank-1 difference 0.0000 everywhere), so the H3 search-mode advantage is not earned at N=256 either - a second falsification, reported as such. Battery pass rates at N=256: 32/256 (1,4-BDO), 60/256 (isobutanol), 22/256 (lycopene). The same amendment’s H2 redirect (Robust Minimal Architecture: minimum heterologous enzyme count K_{het} subject to battery pass, published anchor $K_{\text{het, Yim}} = 5$) is also falsified on pool v3: no battery-passing 1,4-BDO design with $K_{\text{het}} \leq 4$ exists - every robust design requires an upstream route block, so the published route’s enzyme count is not beatable under the robustness constraint in this pool. The remaining live test of the robustness principle is the held-out-target arm (3-HP), locked in the same amendment.

3.7 Objective ablation (amendment 2026-09-27 12:37 IST, locked before computation)

Scoring-component ablation on the committed exhaustive pool-v3 results (the EA itself is inapplicable to ablate at this pool size - the space is enumerated 256/256 - so the operative units are the objective’s components). Four re-rankings were compared per target: full R_{tiered} , nominal-flux only (robustness removed), R_{pert} only, and R_{shift} only. On all three targets all four arms select the *same* winning design; the severe-retention loss of the nominal-only ranking is exactly 0.0. Recorded verbatim per the locked pass rule: the robustness objective is non-binding at pool v3 - robustness and nominal flux are co-maximized by the same architectures on this pool. This is disclosed as a pool property; on larger pools where the trade-off binds, the objective can matter, and the claim made here is only the measured one. Diversity and constraint-penalty ablations are deferred with disclosure (EA-search components, not defined for exhaustive evaluation).

3.8 Unrestricted versus biologically constrained search (amendment 2026-09-27 12:38 IST, locked before computation)

Dropping the growth constraint and maximizing product flux directly over the same pool yields, per target, a measured statement of what the constraint buys. On 14BDO the unrestricted winner reaches 7.75 mmol/gDW/h against the constrained best’s 6.50 (ratio 0.838) and is growth-viable, but it fails the committed battery outright (worst severe-condition retention 0.46): the unconstrained optimum overpromises at nominal and collapses under condition shifts, which is the realism function of the constrained objective in one number. On isobutanol the unrestricted winner is itself battery-robust (0.91), so the constraint is non-binding there; on lycopene the two optima agree within solver tolerance (ratio 1.011, recorded rather than smoothed). No

unrestricted top-ten design is zero-growth on any target. The constraints therefore cost nominal flux only where they also buy measured robustness - a target-dependent answer, reported verbatim.

3.9 What the search is, precisely (queue item 16; definitional pass)

To remove any vagueness about the computational method: the *representation* is a binary genome over six host-lever blocks and two target-route blocks (256 architectures per target); the *search policy* at pool v3 is exhaustive enumeration of that space - the earlier $(\mu + \lambda)$ evolution strategy was retired once enumeration became strictly dominant, so no stochastic search enters any reported number; the *objective* is the locked tiered robustness score $R_{\text{tiered}} = 0.5R_{\text{pert}} + 0.5R_{\text{shift}}$ evaluated by constraint-based flux analysis at 50% condition-specific growth; and the *learning component* is nil - nothing in this pipeline learns from data. It is exact enumeration with a pre-registered scoring rule, and the paper’s claims are limited to what that procedure supports.

3.10 Prioritized experimental shortlist (queue item 17; candidates only, no experiments)

The results yield a ranked, falsifiable validation program. (1) *14BDO constrained-versus-unrestricted A/B*: implement both the R_{tiered} winner (ko_pflB + cofeed_glycerol + route B) and the unrestricted winner (cofeed only, no route block); the framework predicts the unrestricted design retains < 0.5 of nominal production under GLC_LOW/ANAEROBIC while the constrained design retains ≥ 0.8 - a two-strain experiment that can kill the approach. (2) *Isobutanol winner confirmation*: the same architecture class is battery-robust in silico (worst severe 0.91); a single-condition-shifts panel on one constructed strain tests the tiered battery’s severe-leg calibration directly. (3) *Lycopene non-binding check*: constrained and unrestricted optima coincide within tolerance, so lycopene is the negative-control arm - the prediction is no measurable robustness difference. Ordering is by expected information per construct: (1) tests the framework’s central claim, (2) calibrates its battery, (3) controls for over-crediting the constraint.

4 Caveats

(1) All results are in-silico flux predictions; no wet-lab validation is claimed, and no build-phase engineering guidance is produced. (2) Both pools evaluated so far are deliberately small; the exhaustive results bound what these pools can express, and conclusions about search power await larger pools. In particular, the H3 multi-condition advantage is undetectable at pool 64, where the nominal-best and battery-best genomes coincide in every target. (3) The tail-only H2 candidate is verified at the nominal condition only: it fails the tiered battery, every battery-passing design requires the glycerol co-feed, and under equal-feed comparison ($E^{\text{cofeed}} = 1.00$) the co-feed fully accounts for the nominal-flux gains - no flux advantage over the published routes remains. The defensible claims are the robustness structure and its cross-target motif, not nominal flux. (4b) H3’s multi-condition search advantage is falsified at both pool sizes tested (N=64 and N=256): nominal-best and battery-best designs coincide in every target. (4) GSMM predictions assume steady state and fixed biomass composition; real titers are typically lower.

5 Next steps

(1) DONE: co-feed re-baseline ($E^{\text{cofeed}} = 1.00$ on all targets) and the 1000-draw random null (recovers the optimum; nominal search trivial at this size). (2) DONE under the 10:04 amendment: pool-v3 expansion and H2-RMA redirect, both falsified as recorded above. (3) Held-out 3-HP target arm (3B, locked 10:04): construct the malonyl-CoA reductase route, run the identical tiered battery on the 64-genome host-lever pool, Fisher exact test on co-feed + pflB motif enrichment. (4) Yeast8 cross-host arm for the robustness principle.

5.1 Biological realism screen (amendment 2026-09-27 12:32 IST, locked before application)

The top evolved designs were screened against two locked realism filters. F1 (enzyme burden): heterologous reaction count may not exceed the locked literature benchmark pathway's count by more than one; all top-ten designs by R_{tiered} pass on all three targets (benchmarks: 5, 5, and 4 heterologous steps), as do both per-target winners - the evolved architectures are parsimonious relative to demonstrated routes. F3 (cofactor-condition consistency): no top-ten design combines an NADPH-dependent active route block with zero anaerobic retention. F2 (ATP-cost screen) is deferred with explicit disclosure: per-design FBA solution vectors are not stored in the committed results, and computing them is queued as separate compute. No design was re-ranked; the screen is report-only.

5.2 Per-architecture rationale: why the battery winners win

Each battery-winning genome is interpretable block by block, and the interpretation is the same logic the tiered battery was designed to select for. **1,4-BDO winner** (ko_pflB + glycerol co-feed + succinyl-CoA route, nominal 6.496 mmol/gDW/h, $R_{\text{tiered}} = 0.994$): the pflB knockout removes the formate overflow arm, keeping pyruvate carbon inside the central network where the succinyl-CoA route can draw on it; the co-feed adds a second, more reduced carbon source whose uptake is uncoupled from glucose repression in the model. The design trades nothing at nominal and retains frontier-relative flux on every severe shift except anaerobiosis (0.938), where formate removal is nearly free because pyruvate formate-lyase is itself oxygen-sensitive. **Isobutanol winner** (ko_pflB + ko_adhE + co-feed + NADH-alcohol dehydrogenase, 6.498, $R_{\text{tiered}} = 0.9997$): the adhE knockout blocks the ethanol drain on acetyl-CoA and NADH, and choosing the NADH-dependent alcohol dehydrogenase matches the cofactor pool the knock-offs enrich - the architecture is internally redox-consistent, which is exactly the property the mild oxygen-limitation leg (retention 1.0) measures. **Lycopene winner** (co-feed + idi push, 0.730, $R_{\text{tiered}} = 1.0$, the only perfect score): the idi push balances IPP:DMAPP isomer supply - the known bottleneck of the heterologous crtEBI tail - while the co-feed raises precursor flux without touching dxs; the design reaches the battery frontier in all nine conditions, including anaerobiosis, where its retention is 1.0. Common logic across all three: carbon supply is the binding lever - *every* battery-passing genome in all three targets carries the glycerol co-feed (32/32, 60/60, and 22/22 of passers), while no genome passes without it, and conservation knockouts are optional score-improving refinements rather than requirements (co-feed-only genomes pass in all three targets; the rank-1 designs pair the co-feed with a knockout in two of three). The design principle the battery surfaces at N=256 is therefore sharper than a generic conjunction: unlock carbon supply first, then conserve it. This is a statement about which levers

robust designs share, not about a multi-condition search advantage - H3 was falsified at $N=256$ (argmax coincidence, above) and nothing here resurrects it.

5.3 Held-out target: 3-hydroxypropionate (3B)

The remaining live test of the robustness principle was run under its 10:04 IST lock: 3-HP via the heterologous MCR1/MCR2 route in the same iML1515 base, 64 host-lever genomes, identical tiered battery with the severe frontier computed within the 3-HP pool. Six of 64 genomes pass both hard legs (nominal max 10.91 mmol/gDW/h); none carries the locked motif $M = \{\text{co-feed, ko_pflB}\}$ (Fisher one-tailed $p = 1.0$), so H0,3B is not rejected and the specific winner architecture is falsified as a portable motif, per the locked rule with no loosening. The motif's decomposition is the finding: all six passers carry the glycerol co-feed (the supply-lever necessity replicates on a held-out chemistry), while none carries the pflB knockout (PFL carries anaerobic pyruvate flux the NADPH-demanding MCR route cannot spare). The transferable principle is carbon-supply unlock; knockout pairings are target-specific.

5.4 FBA-assumption sensitivity (queue #4)

The three battery winners were re-evaluated on a locked 12-cell grid (biomass fraction $\{0.4, 0.5, 0.6\} \times$ glycerol uptake $\{-2, -4\} \times$ glucose base $\{-8, -10\}$), surviving 6/12 (1,4-BDO), 8/12 (isobutanol), and 10/12 (lycopene) cells. The mild leg never fails (minimum mild retention 1.0 in all 36 cells); all failures sit in the stacked worst case (forced growth 0.6 with throttled feeds) on the severe leg. Robustness therefore holds under single-parameter perturbation and degrades gracefully, with the lycopene design most assumption-stable.

5.5 Monte Carlo FBA uncertainty (queue #15)

Under the 13:35 IST lock, each winner was re-evaluated over 200 draws with every battery uptake bound independently jittered by $U(0.8, 1.2)$ (seed 260927). Battery-pass fractions: 1,4-BDO 92.5% (mean $R_{\text{tiered}} = 0.993$, 5–95% $[0.949, 1.043]$), isobutanol 98.0% (0.999, $[0.955, 1.049]$), lycopene 100% (0.999, $[0.965, 1.037]$). Winner robustness is not a knife-edge property of the locked bounds; the 7.5% 1,4-BDO failure tail is reported as the sensitivity mass below the pass bar.

5.6 Cross-organism transfer: native-yeast arm (queue #11)

The carbon-supply principle was tested in Yeast8 (native succinate export, 50% growth, 7 conditions mirroring the battery; amendment 14:15 IST, all reaction IDs verified pre-lock). The glycerol co-feed improves succinate flux in 6 of 7 conditions with the largest gain under carbon limitation (GLC_LOW 1.49 $\rightarrow$ 3.05 mmol/gDW/h, +104%) - but collapses anaerobically (6.81 $\rightarrow$ 0.00), consistent with yeast's oxygen-dependent glycerol catabolism, so the locked all-conditions prediction is reported as not met. Together with the 3-HP hold-out, the transferable content is the abstract principle (unlock carbon supply, gains largest under limitation); lever choice is host-specific.

5.7 Fourth chemical class: native succinate (queue #10)

A native organic-acid target was added under the 14:17 IST lock (64 host-lever genomes, identical battery). Ten of 64 genomes pass both hard legs (nominal max 10.66 mmol/gDW/h), and all ten carry the glycerol co-feed (Fisher one-tailed $p = 0.0004$, pre-locked test): the carbon-supply-necessity pattern replicates on a fourth, chemically distinct target class, its first replication under a pre-registered significance gate.